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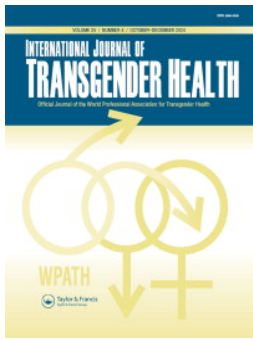


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The unique roles of family, friend, and online support in the psychological adjustment of transgender young adults

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ABSTRACT

Background: Transgender youth are more likely than cisgender youth to encounter risk, such as a lack of social support from families and offline friends, leading many to turn to online friendships for social support. Feeling supported is crucial for psychological adjustment, measured in positive outcomes such as pride, self-esteem, and belongingness and in negative outcomes such as internalized transphobia, depression, and loneliness.

Aims: Using the risk and resilience framework, this study explores the relationship between social support across social contexts (families, offline friends, and online friends) and psychological adjustment using hierarchical regression modeling in a sample of 335 binary and nonbinary transgender young adults.

Results: Family and offline friend support had stronger associations with adjustment than online friend support, and online friend support moderated the relationship between family support and pride and the relationship between offline friend support and pride. A qualitative analysis of participants' experiences of being transgender online explored additional insights into how transgender young adults use the internet for support. The qualitative analyses contextualized the qualitative findings, showcasing the varied experiences of trans individuals online and the importance of offline support as a cornerstone for psychological adjustment.

Conclusions: Social support from offline social contexts are more foundational and paramount to promoting positive psychological adjustment than online social support. Online engagement in trans young adults is individualized and offers a variety of supportive and unsupportive experiences.

KEYWORDS

Nonbinary; risk and resilience; social support; transgender; transgender adjustment

Introduction

Recent years have been characterized by an increase in public anti-LGBTQ sentiment across the globe (International Lesbian, Gay, Bisexual, Trans, and Intersex Association [ILGA], 2024; The Trevor Project, 2023), and leading to an increase in anti-LGBTQ legislative bills proposed globally in countries such as the United States, India, Russia, and Uganda (American Civil Liberties Union [ACLU], 2023; Evdokimova, 2023). Bills targeting trans youth range from limiting access of trans youth to medical care, barring transgender individuals from using public bathrooms, forcing teachers to out trans students to parents, and decreasing nondiscrimination protections which prohibit employers, hospitals, and businesses from refusing to work with LGBTQ individuals (Evdokimova, 2023; ACLU,

2023). While there is a global trend of rising LGBTQ affirmative policy over the past 20 years, legislation offering protections for transgender individuals are still rare globally (Williamson, 2024). With limited legal protections in place, the recent wave of anti-trans legislation limits the ability of trans and nonbinary youth to receive social and medical support.

In a 2023 poll of over 700 American LGBTQ youth, 89% of trans youth reported that their mental health was negatively affected by anti-trans legislation. 30% of LGBTQ youth reported facing online harassment based in anti-LGBTQ policies, and trans youth make up 45% of those reporting such harassment. In the wake of health care bans, 29% of trans youth do not feel comfortable seeking medical care when sick or injured, a rate 7.25 times higher than for non-trans LGBTQ youth.

Health care bans and school policies that out LGBTQ students to parents have left trans youth feeling intensely fearful and report that political issues of transphobia heavily contribute to their daily stress and anxiety levels (The Trevor Project, 2023). Clearly, trans youth face unprecedented threats to their well-being.

The risk and resilience perspective provides an important framework for understanding the challenges faced by trans youth. According to the risk and resilience framework, trans youth are more likely to experience systemic adversity than their cisgender peers (Eisenberg et al., 2017; Johns et al., 2019). In the context of this framework, risks impede normative developmental processes through exposure to threats and adversities and often lead to maladaptive outcomes such as depression and suicide (Arrington & Wilson, 2000; Catterall, 1998; Luthar et al., 2000). Heightened levels of risk are present in virtually every context in which trans youth develop (e.g., at home, at school, in the community), and they threaten trans youths' physical, social, and emotional well-being (Stieglitz, 2010; Tankersley et al., 2021). For example, trans youth who disclose their gender identity face discrimination and victimization by parents, peers, and employers (Singh et al., 2014; Stieglitz, 2010). These experiences, in turn, contribute to significant negative physical and mental health outcomes. Trans youth experience higher rates of substance use, risky sexual behaviors, emotional distress, and bullying victimization than their cisgender peers (Eisenberg et al., 2017; Johns et al., 2019). Research has found high rates of suicidal ideation and suicide attempts in trans youth, with between 44% and 82% of trans youth reporting suicidal ideation, and between 31% and 40% of trans youth attempting suicide (Austin et al., 2022; Eisenberg et al., 2017; Johns et al., 2019; Veale et al., 2017).

Resilience is the ability of an individual faced with risks to overcome or reduce the negative effects brought about by risk exposure through the presence of factors that bring about positive outcomes, reduce or eliminate negative outcomes, or both (Fergus & Zimmerman, 2005). Resilience factors can be qualities of the individual person or of the environments in which they are

situated, and while it is easy to conceptualize resilience as an extraordinary feat, most are ordinary qualities, such as finding community and promoting belongingness (Masten, 2001). Resilience is multidimensional and contextual, so it is possible to be resilient in one context and not another, such as finding success despite risk exposure in a school environment, but not in a risky familial setting (Fergus & Zimmerman, 2005; Luthar et al., 2000; Southwick et al., 2014). Researchers have identified important sources of resilience among trans youth. For example, having a strong sense of community with and support from other trans individuals is one of the most effective resilience factors, and engagement with other transgender individuals predicts less fearfulness and suicidality and a stronger sense of comfort (Singh et al., 2011; Tankersley et al., 2021; Testa et al., 2014).

The current study focuses on the associations of social support from family, friends, and online communities with indicators of social and emotional adjustment and well-being among transgender and nonbinary young adults. We conceptualized social support as a source of both risk (when levels of support are low) and resilience (when levels are high), with a focus on how support from online connections and communities might moderate the associations between support from family and "real life" friends and adjustment indicators like depression, anxiety, and self-esteem. Trans youth are less likely to receive social support from family than cisgender youth, and this threatens their well-being and mental health (Alanko & Lund, 2020; Davey et al., 2014; Dowers et al., 2020). At the same time, while low levels of social support are a risk factor, high levels of support can serve as a resilience factor (Fergus & Zimmerman, 2005). We were particularly interested in whether social support from online sources might serve as a buffer against the harmful outcomes associated with low family and peer support.

Parental and peer support

Social support taps into the basic human need to belong, and receiving social support is associated with both positive physical and mental health

outcomes for both trans youth and cisgender youth (Alanko & Lund, 2020; Baumeister & Leary, 1995; Hefner & Eisenberg, 2009; Hale et al., 2005). Social support from trans-affirming parents and peers can provide resilience to trans youth, who report lower levels of social support than LGB youth (Başar & Öz, 2016; Ryan et al., 2010). Support from parents is also associated with a lower risk of depression and suicidality (Bosse et al., 2024; Boza & Nicholson Perry, 2014; Milton & Knutson, 2023; Moran et al., 2018; Simons et al., 2013), and a strong connection with at least one parent provides a 23% reduction in the likelihood of developing depressive symptoms (Singh et al., 2014; Singh & McKleroy, 2011; Tankersley et al., 2021). Additionally, high parental support in adolescence predicts positive life adjustment in young adulthood *via* high self-esteem, pride in their LGBTQ identity, and life satisfaction (Ryan et al., 2010; Snapp et al., 2015). Supportive families also help reduce internalized transphobia in trans youth by helping them feel less embarrassed for expressing their transgender identity (Scandurra et al., 2017; Simons et al., 2013). Low parental support is associated with an increased risk of personal distress, depression, substance abuse, non-suicidal self-injury, risky sexual behavior, and suicidality (Claes et al., 2015; McConnell et al., 2016; Moran et al., 2018; Ryan et al., 2010; Wilson et al., 2012). Trans adolescents who receive little support from parents are also at a higher risk for psychological distress in adulthood (McConnell et al., 2016). If LGBTQ youth are not supported by their family, they may choose to instead adopt a “chosen family” of peers for guidance, support, and resources. While support from both families of origin and chosen families are effective in reducing the risk of depression, support from chosen families does not account for much change in depression compared to the support of their families of origin (Milton & Knutson, 2023).

The qualitative literature also highlights the importance trans youth place on peer and community support (Kia et al., 2021). Peer support from other trans youth in particular helps normalize the trans experience and turns shame for their identity into pride. Additionally, this

support helps trans youth navigate trans-specific issues, like medical and social transitioning, and validates their identity, empowering them for a better future (Johnson & Rogers, 2020). Some literature finds that parental support is more effective than peer support in reducing suicide risk in trans populations (Austin et al., 2022; Bauer et al., 2015); however, there is also evidence that peer support is a more effective buffer against depression and victimization than family support (Moran et al., 2018), and that both family and peer support are effective in promoting positive mental health (Başar et al., 2016; Snapp et al., 2015). Low peer support serves as a risk factor for trans youth who often face ostracization from peers, leading to social and emotional isolation (Singh et al., 2014; Tankersley et al., 2021). Trans youth are vulnerable to victimization from unsupportive family and peers in the form of harassment, bullying, disownment, violence, and sexual objectification, putting them at risk for developing low self-esteem, depression, substance abuse, and suicidality (Eisenberg et al., 2017; Stieglitz, 2010). Feeling supported by one's family, peers, and community is paramount for trans resilience.

Online support for transgender youth

A sense of connectedness is important for trans youth, and when they cannot find community in person, they may turn to the internet for online community support (Austin et al., 2020; Rothbaum et al., 2022; Selkie et al., 2020). LGBTQ youth seek affirming spaces, and 68% of American LGBTQ youth view online social spaces as affirming while viewing other offline social domains, such as school (54%) and home (38%), as less likely to be affirming (The Trevor Project, 2023). Engagement in online communities for social support is associated with lower loneliness and depression and higher self-esteem, satisfaction with life, sense of community, and felt support (Oh et al., 2014; Shaw & Gant, 2002). LGBTQ youth are more likely than non-LGBTQ youth to have online friendships and value online friendships over offline friendships (Ybarra et al., 2015). LGBTQ youth use the internet in many of the same ways as those not in the LGBTQ community, including seeking friendships, education,

and entertainment; however, LGBTQ youth also use the internet in unique ways that bolster their sense of community with other LGBTQ youth (Craig et al., 2015; Craig et al., 2021). For example, trans youth often seek online communities of other trans individuals with whom they can relate. In these communities, trans youth can find social and emotional support, positive trans role models, information on medical decisions and medical transition, advice for explaining trans issues to cis family and peers, support against victimization, and a safe space to log and share gender transition milestones (Adkins et al., 2018; Craig et al., 2021; Rothbaum et al., 2022; Selkie et al., 2020; Steinke et al., 2017).

Current study

Transgender youth face unique limits to their ability to receive social support and find community. A lack of social support is associated with poor mental health outcomes, while social support, specifically from family, peers, and communities of trans individuals, is important to resilience in trans populations (Alanko & Lund, 2020; Eisenberg et al., 2017). Because many trans youths do not live in supportive environments, many seek friendship and community online (Austin et al., 2020; Selkie et al., 2020; Steinke et al., 2017; Ybarra et al., 2015). While past research on LGBTQ youth has compared offline and online levels of social support and differences in support from family and friends, no study has investigated the experiences of family, offline friend, and online friend support in trans youth, specifically in a historical context where trans youth's belongingness needs are threatened by record levels of anti-trans legislation. The current study took a comprehensive approach, assessing how support from family, offline friends, and online people or communities link to an array of indicators of health and well-being among trans and nonbinary young adults. Further, we were interested in the potential moderating effect of online support in the associations between family and friend support and adjustment.

Three specific hypotheses were tested. First, we expected that social support would have a positive association with self-esteem, belongingness,

pride, satisfaction with life, and subjective well-being, and a negative association with loneliness, depression, anxiety, and internalized transphobia, across all three social contexts. Second, we predicted that offline social support would be more effective in promoting positive adjustment outcomes and mitigating negative adjustment outcomes than online social support. Finally, we predicted that online support would moderate the relationship between offline support (both family and offline friends) and adjustment indicators (both positive and negative). Specifically, we anticipated that online social support would promote even more positive adjustment for individuals who already experience social support from others, and buffer against negative adjustment for those who do not receive offline social support.

As an additional mode of exploration of the potential importance of online support for trans individuals, we also conducted a thematic analysis of participants' open-ended responses to a question about the role of the internet in their experiences as a trans person. This additional analysis allows for a more nuanced understanding of the role of online support in the lives of trans individuals and supplements the quantitative analyses with the lived experiences of trans individuals who use the internet.

This study advances our understanding of the risks and resilience factors experienced by trans and nonbinary young adults in numerous ways. First, this study investigates the relationship between support and adjustment across multiple contexts (family, offline friends, and online friends). While previous research has compared family vs. friend contexts (e.g., Milton & Knutson, 2023) and online vs. offline contexts (e.g., Ybarra et al., 2015), none has compared levels of support across all three contexts. This is especially unique, as this allows for comparisons of low and high levels of support across multiple contexts and how these interactions predict adjustment. This allows for a nuanced investigation into the potential buffering role of online support in the link between low "real-life" support and psychological adjustment. Additionally, since empirical psychology is often criticized for centering research on sexual and gender minorities' negative experiences instead of research on both positive and

negative experiences (e.g., Levitt et al., 2023), this study investigates the complex role of support in predicting both positive and negative adjustment. Finally, this study serves as a unique historical opportunity to study the experiences of trans young adults in an age of heightened public and political transphobia which places increased pressure on trans individuals (ACLU, 2023; The Trevor Project, 2023).

Methods

Participants

This preregistered study used data from 335 participants ($M. age = 21.99$, $SD = 2.031$) who identified across a spectrum of transgender identities: binary trans men (23.3%), binary trans women (13.4%), nonbinary (45.4%), agender (4.2%), questioning (1.2%), and a multitude of nonbinary identities such as genderqueer, genderfluid, and demigender (11.6%). Participants were recruited using Prolific, a service that connects researchers and participants based on demographic criteria. Transgender and nonbinary young adults aged 18 to 25 living in the United States were eligible for participation. While this sample featured a diverse collection of transgender identities, it was not particularly racially or ethnically diverse; 72.4% of participants were white, 8.41% were black, 5.11% were Asian, 2.7% were Indigenous American, 0.3% were Hawaiian or Pacific Islander, and 11.11% were of another racial background not listed. The majority of participants were neither Hispanic nor Latino (83.5%).

After providing online consent, participants completed an online survey *via* the Qualtrics platform that was designed to take approximately 20 min. Participants were compensated at an average rate of \$12.91 per hour, and the median completion time was 19 min. This study was approved by the University of Oklahoma Institutional Review Board.

Measures

Friendships

Participants were asked to indicate the number of friends with whom they interact exclusively

online, exclusively offline, and both online and offline.

Social support

Social support from offline friends, online friends, and family was measured using the *family support* and *friend support* subscales of Zimet et al.'s Multidimensional Scale of Perceived Social Support (1988). In addition to the traditional *friend support* subscale from the MSPSS, we also included a modified version of that subscale to tap into perceived support from online friends. Each subscale included five items that asked participants to rate their agreement on a scale from 1 (strongly disagree) to 7 (strongly agree). An example item from this scale is, "I can count on my family/friends/online friends when things go wrong." All three subscales showed excellent internal consistency: family ($\alpha = 0.93$), offline friends ($\alpha = 0.95$), and online friends ($\alpha = 0.93$).

Positive adjustment indicators

Self-esteem was measured using the 10-item Rosenberg Self-Esteem scale (1965; $\alpha = 0.92$). This measure asked participants to rate their agreement with each item on a scale from 1 (strongly disagree) to 5 (strongly agree), and an example item from this scale is, "I feel that I have a number of good qualities." **Belongingness** was measured using Malone et al.'s 12-item General Belongingness scale (2012; $\alpha = 0.95$). This scale asked participants to rate their agreement with each item on a scale from 1 (strongly disagree) to 7 (strongly agree), and an example item from this scale is, "When I am with other people, I feel included." **Pride in transgender identity** was measured using the seven-item *pride* subscale of Testa et al.'s Gender Minority Stress and Resilience Measure (2015; $\alpha = 0.87$). This scale asked participants to rate their agreement with each item on a scale from 1 (strongly disagree) to 5 (strongly agree), and an example item from this scale is, "My gender identity or expression makes me feel special or unique." **Satisfaction with life** was measured using Diener et al.'s five-item scale (1985; $\alpha = 0.91$). This scale asked participants to rate their agreement with each item on a scale from 1 (strongly disagree) to 5 (strongly agree),

and an example item from this scale is, “In most ways my life is close to my ideal.” **Subjective well-being** was measured using the 5-item World Health Organization’s Well-Being Index ($\alpha=0.88$; Topp et al., 2015). This scale asked participants to rate their agreement with each item on a scale from 1 (at no time) to 6 (all of the time) over the past 2 wk, and an example item from this scale is, “I have felt cheerful and in good spirits.”

Negative adjustment indicators

Loneliness was measured using the 10-item UCLA Loneliness scale ($\alpha=0.91$; Russell et al., 1978). This scale asked participants to rate their agreement with each item on a scale from 1 (never) to 4 (often), and an example item from this scale is, “How often do you feel as if no one understands you?” **Depression** was measured using the 12-item Beck Depression Inventory ($\alpha=0.92$; Beck & Beck, 1972). This scale asked participants to rate the intensity of their feelings using four statements increasing in intensity. An example from this inventory is, “I do not feel sad,” “I feel sad,” “I am sad all of the time and can’t snap out of it,” and “I am so sad or unhappy that I can’t stand it.” For each item, participants identified which of the four statements most accurately described them. Each statement is scored on a scale from 0 to 3, where 0 represented no symptoms, while 3 represented high severity. The mean of all items was used in analyses.

Anxiety was measured using the 5-item trait anxiety subscale of the short version of the Spielberger State-Trait Anxiety Inventory ($\alpha=0.85$; Zsido et al., 2020). This scale asked participants to rate their agreement with each item on a scale from 1 (not at all) to 4 (very much so), and an example item from this scale is, “I get in a state of tension or turmoil as I think over my recent concerns and interests.” **Internalized transphobia** was measured using the internalized transphobia subscale of Testa et al.’s Gender Minority Stress and Resilience Measure (2015; $\alpha=0.92$). This scale asked participants to rate their agreement with each item on a scale from 1 (strongly disagree) to 5 (strongly agree), and an example item from this scale is, “I feel that my gender identity or expression is embarrassing.”

Qualitative Measure of internet experiences

To further investigate the nuanced experiences of trans young adults using the internet, participants were presented with an optional open-ended prompt which read, “Briefly describe how your time on the internet has influenced your life as a trans/nonbinary young adult.”

Analytic approach

Data were anonymized and cleaned, and the final data set excluded participants that did not complete the survey or pass the attention checks. To test the hypotheses, first, a series of correlational analyses were conducted to test the association between social support across social contexts and the adjustment variables. Additionally, a series of Fisher’s r -to- z tests were conducted to further clarify the relative magnitude of the associations between support and adjustment.

To further test the hypotheses, a series of hierarchical linear regressions was conducted to explore the relationship between each context of social support and adjustment in greater depth. Each model comprised four steps and set an index of adjustment as its outcome variable (see Table 2). Step 1 included the number of offline and online friends as covariates. Step 2 included social support from families and offline friends as predictors of the respective adjustment outcome. Step 3 added social support from online friends as a predictor. Finally, Step 4 included two interactions between predictor variables: one between family support x online friend support and the other between offline friend support x online friend support. All variables were centered prior to analysis. Significant interaction terms were further probed using prototypical plots as described by Aiken and West (1991). This procedure involves estimating the association between family or friend support and adjustment at high and low levels of online peer support, based on values at one standard deviation above and below the mean.

Participants were given the option to provide a written response to the following prompt, “Briefly describe how your time on the internet has influenced your life as a trans/nonbinary young adult.”

Of the 335 participants who completed the quantitative portion of the study, 206 (over 60%) completed the open-ended item. Data were coded holistically based on the content discussed in each statement. Of particular interest were references to supportive interactions and/or unsupportive interactions from family, offline friends and community, and online friends and community, for a total of 618 codes. Two independent coders noted whether each response referenced each theme of interest with a 91% agreement rate. All disagreements between coders were resolved through discussion.

Results

Correlational analyses

A series of correlational analyses was conducted to explore the relationship between social support from family, offline friends, and online friends and all adjustment variables (see Table 1). Social support from families was positively correlated with pride, belongingness, self-esteem, satisfaction with life, and positive well-being (r s .19 to .55, $ps < .01$) and negatively correlated with loneliness, trait anxiety, depression, and internalized transphobia (r s -0.17 to -0.42 , $ps < .01$). Support from offline friends was also positively correlated with pride, belongingness, self-esteem, satisfaction with life, and positive well-being (r s .19 to .55, $ps < .01$) and negatively correlated with loneliness,

trait anxiety, depression, and internalized transphobia (r s -0.18 to -0.44 , $ps < .01$). Conversely, support from online friends was only modestly positively correlated with pride, belongingness, and positive well-being, and negatively correlated with loneliness. The strongest relationships for support and adjustment across each social context were between support and belongingness. Additionally, while family and offline support were significantly associated with each other, $r(327) = .38$, $p < .001$, online friend support was only weakly associated with offline friend support, $r(327) = .12$, $p = .025$, and not associated with family support.

A series of Fisher's r -to- z tests further clarified the relative magnitude of the associations between support and adjustment. We found no significant differences between the correlations of family and offline friend support with adjustment. However, significant differences were found between correlations with family support and online friend support, and between offline friend support and online friend support and all adjustment outcomes except pride. This indicates that sources of offline support are more strongly associated with psychological adjustment than online support.

Regression analyses Predicting adjustment from social support

A series of hierarchical linear regressions was conducted to further explore the relationship

Table 1. Summary of correlations between social support and adjustment outcomes.

	Pride	SE	Belong	SWL	PWB	IT	Lone	Dep	Anx
Family Support	.187**	.452**	.553**	.460**	.431**	-.167**	-.422**	-.401**	-.292**
Offline Friend Support	.191**	.404**	.551**	.385**	.35**	-.179**	-.435**	-.333**	-.208**
Online Friend Support	.162**	.069	.200**	.074	.153**	-.063	-.172**	-.065	.039

SE = Self Esteem; Belong = Belongingness; SWL = Satisfaction with Life; PWB = Positive Well-Being; IT = Internalized Transphobia; Lone = Loneliness; Dep = Depression; Anx = Anxiety. **Correlation is significant at the 0.01 level (2-tailed).

Table 2. Summary of beta weights from hierarchical regressions predicting adjustment outcomes from family, offline friend, and online friend support.

	Pride	SE	Belong	SWL	PWB	IT	Lone	Dep	Anx
Total Adjusted ΔR^2 :	.09	.30	.47	.29	.24	.06	.29	.21	.09
Step 1:									
Online Friends	ns	ns	.122	ns	ns	ns	-.136	ns	ns
Offline Friends	.142	.332	.265	.335	.209	-.148	-.213	-.233	-.132
Step 2:									
Family Support	.127	.327	.389	.343	.337	ns	-.290	-.305	-.245
Offline Friend Support	.123	.219	.382	.186	.184	ns	-.306	-.177	ns
Step 3:									
Online Friend Support	.139	ns	.117	ns	ns	ns	ns	ns	ns
Step 4:									
Family x Online Support	.152	ns	ns	ns	ns	ns	ns	ns	ns
Offline x Online Support	-.161	ns	ns	ns	ns	.168	ns	ns	ns

SE = Self Esteem; Belong = Belongingness; SWL = Satisfaction with Life; PWB = Positive Well-Being; IT = Internalized Transphobia; Lone = Loneliness; Dep = Depression; Anx = Anxiety.

between each context of social support and adjustment. There were only significant interaction terms in the hierarchical regressions predicting pride and internalized transphobia. Below, we first describe the results from these two analyses, and then turn to summarizing the results from the remaining regressions.

The hierarchal regression model using pride as the outcome found that the number of offline friends ($\beta = .028, t=2.56, p = .011$), family ($\beta = .071, t=2.18, p = .03$), offline friend ($\beta = .072, t=2.03, p = .043$), and online friend ($\beta = .092, t=2.43, p = .016$) support significantly predicted pride (see Table 2). Additionally, both the interactions between family x online support ($\beta = .055, t=2.52, p = .012$) and offline x online support ($\beta = -0.053, t=-2.66, p = .008$) significantly predicted pride. Prototypical plots revealed a positive association between family support and pride at low levels of online friend support ($\beta = .028, t=6.57, p < .001$) and a stronger positive association between family support and pride at higher levels of online friend support ($\beta = .043, t=9.03, p < .001$, see Figure 1). Online friend support also moderated the relationship between family support and pride. Prototypical plots revealed a negative association between offline friend support and pride at low levels of online support ($\beta = -0.142, t=-4.39, p < .001$), and a weaker association between offline friend support and pride at higher levels of online support ($\beta = -0.029, t=-9.00, p < .001$, see Figure 2).

A hierarchical regression using internalized transphobia as the outcome variable found that only the

number of offline friends ($\beta = -0.033, t=2.56, p = .011$) and the interaction between offline x online support ($\beta = .064, t=2.73, p = .007$) significantly predicted internalized transphobia. Prototypical plots did not find significant differences between slopes in the interaction of offline and online friend support predicting internalized transphobia.

Out of all other hierarchical regressions, only one (for belongingness) found that online friend support predicted adjustment outcomes over and above the variance explained by the covariates and family and offline friend support.

The remaining models found varying relationships between the predictor variables and outcomes. The number of offline and online friends (Step 1) and support from families and offline friends (Step 2) significantly predicted lower loneliness. A greater number of offline friends (Step 1) and more perceived support from families and offline friends (Step 2) significantly predicted higher self-esteem, satisfaction with life, and positive well-being, and fewer depressive symptoms. The number of offline friends (Step 1) and greater support from families (Step 2) significantly predicted less trait anxiety. A summary of these regression analyses can be found in Table 2, and a summary of R-square changes across each step can be found in Table 3. While support from online friends is a predictor of some adjustment outcomes (pride, internalized transphobia, and belongingness), support from families and offline friends were stronger predictors of psychological adjustment as measured by the regression weights and variance explained.

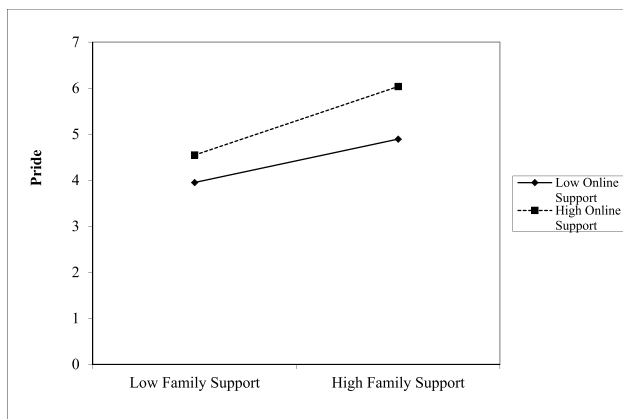


Figure 1. Online friend support moderates the relationship between family support and pride.

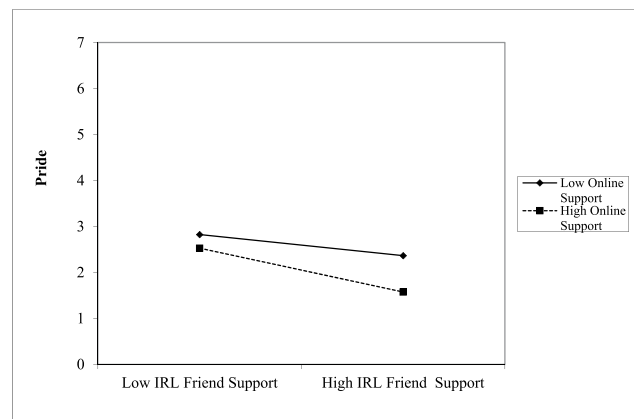


Figure 2. Online friend support moderates the relationship between offline friend support and pride.

Additional analyses

When first reviewing the data, we noted that when reporting the number of in-person, online, and hybrid in-person/online friends, 40.5% of the sample reported having no in-person friends. A multivariate ANOVA was conducted to determine whether having offline friendships (coded in a yes/no binary) was associated with adjustment outcomes (see Table 4). Participants with offline friends reported better adjustment than those without offline friends for all indicators except pride and anxiety, with small to medium effect sizes.

Qualitative analysis of online support

Of the 206 responses to the optional prompt, “Briefly describe how your time on the internet has influenced your life as a trans/nonbinary young adult,” 139 (67%) contained comments related to these themes. Thirteen participants (9%) mentioned family, and of these cases, all described unsupportive family relationships. Fifty-three (38%) mentioned offline friends and communities, and of these, 74% indicated unsupportive interactions, 19% indicated supportive interactions, and 8% indicated mixed support. One hundred twenty-nine participants (93%) referenced online friends and communities, and of these, 8% indicated unsupportive interactions, 74% indicated supportive interactions, and 18% mixed support.

The predominance of comments related to online interactions compared to family and offline interactions was expected given the directions in the prompt itself (which focused on online experiences); however, below we present a thematic analysis of all responses related to support.

Supportive experiences online

In those who discussed unsupportive families, most indicated growing up in conservative religious households and communities who were unsupportive of queer identities. Some indicated only expressing their identities online out of fear of familial rejection, while others indicated that involvement in online communities allowed them to gain the confidence and knowledge to come out to their families.

We found common themes among those who discussed support from online friends and communities. Many, especially those who grew up in conservative religious families, discussed that they only discovered the existence of the LGBTQ community *via* the internet. Because they did not encounter other out queer people in their offline lives, they only discovered queer identities through the internet and online communities. Engaging in online communities with other trans individuals was important for many, with one person saying, “I feel encouraged when I see the posts and comments of other trans/nonbinary

Table 3. Summary of R-squared change from hierarchical regressions Predicting adjustment outcomes from family, offline friend, and online friend support.

	Pride	SE	Belong	SWL	PWB	IT	Lone	Dep	Anx
Step 1 ΔR^2 Change	.03*	.11*	.09*	.12*	.06*	.03	.07*	.06*	.02
Step 2 ΔR^2 Change	.04*	.19*	.37*	.19*	.18*	.03*	.22*	.15*	.09*
Step 3 ΔR^2 Change	.02	.00	.01*	.00	.01	.00	.01	.00	.01
Step 4 ΔR^2 Change	.03*	.01	.00	.01	.00	.03	.01	.01	.00

*R-squared change is significant at the .01 level. SE=Self Esteem; Belong=Belongingness; SWL=Satisfaction with Life; PWB=Positive Well-Being; IT=Internalized Transphobia; Lone=Loneliness; Dep=Depression; Anx=Anxiety.

Table 4. Summary of multivariate analysis predicting adjustment outcomes from having offline friendships.

	Offline friendships <i>M</i> (<i>SE</i>)		<i>F</i>	Sig.	Partial eta squared
	No	Yes			
Pride	3.008 (.083)	3.108 (.069)	.867	.353	.003
SE	2.483 (.081)	2.903 (.067)	16.002	<.001	.047
SWL	2.135 (.096)	2.613 (.08)	14.584	<.001	.043
PWB	2.574 (.087)	2.83 (.073)	5.068	.025	.015
Belong	3.65 (.117)	4.101 (.098)	8.725	.003	.026
IT	2.403 (.096)	2.268 (.08)	1.163	.282	.004
Lone	3.078 (.061)	2.912 (.051)	4.438	.036	.014
Dep	2.238 (.062)	2.049 (.052)	5.497	.02	.017
Anx	2.823 (.069)	2.662 (.057)	3.234	.073	.01

SE=Self Esteem; Belong=Belongingness; SWL=Satisfaction with Life; PWB=Positive Well-Being; IT=Internalized Transphobia; Lone=Loneliness; Dep=Depression; Anx=Anxiety.

folks, and I feel like part of that online community. I think I would be questioning if anything I feel about my identity is real if I didn't see my experiences echoed in other peoples' anecdotes."

Support from online friends and communities helped bolster their self-acceptance, confidence, and understanding by finding others who could relate to them based on their identities and experiences. These relationships also helped many reduce their internalized transphobia by normalizing trans existence. Some commented on the importance of seeing trans joy, others on living happy euphoric lives full of meaning and purpose. One person commented:

I think the biggest influence for me has been seeing other people be out and proud in their identity and gaining inspiration from their courage. I am in a lot of social media groups dedicated to trans joy and expression..., and I get a lot of meaning from seeing that I'm not alone. I enjoy seeing other people's journeys play out and seeing how those journeys are similar/different to mine. It reduces my feelings of isolation even though I haven't made any "friends" in these groups.

Support from online communities took many forms, sometimes in the form of social/emotional support from trans communities, while other times it was through informational support, gaining advice and insights on things like social and medical transitioning, how to access gender affirming care, navigating legal processes for changing government IDs, and gaining fashion inspiration to better express their identities. Many who lacked support offline described online spaces as a "beacon of hope" or a "light in the darkness," cultivating a sense of hope and purpose, and in some cases, giving them a reason to continue living. One person recalled, "It's been the only thing that kept me alive really. [The] only place I could feel somewhat like I'm not a freak and that there are people who are like me or who accept me."

Those who received support from offline friends and communities reported a variety of ways in which the internet influenced them as a trans person. Some saw the internet as a helpful space for learning about and developing their own identities but found that online connections were best used to augment, rather than to replace, offline ones, with one person commenting:

It is easier to find community online than it is in real life, but I still strive for more in-real life connections and relationships. I do feel that it has played a large role in helping me discover and unpack many things I feel in regard to my sexual and gender identity, and while I appreciate and value those lessons, I do not want it to replace relationships I can establish with people in my offline life.

Expressing similar ideas, another commented:

Growing up I lived pretty far from most of my friends. My best friend and I went to school together but lived in different cities, so we primarily talked online. For me, this has always been the baseline of interacting—online is an augmentation for in person. Obviously now I have friends I've met only online, and I would say the main way that interacts with my existence as a queer person is that it's the only space where I actively take part in things and seek people out purely on the basis of queerness.

Still, others are entirely reliant on online communities to express their gender identities, with one person saying:

[Without the internet,] I would not know that my gender identity could present as it does or could mean what it does to me if I had not had access to the internet and other people that feel like I do. I worry a lot of the time that I have taken too much influence from the internet, but I also recognize that the alternative is just feeling unsettled and different without knowing why. I feel more at home on the internet and feel disconnected between the online and offline worlds, because my gender identity is essentially irrelevant in my offline life. I feel like people in my offline life do not care about that level of my identity personally, and they do not ask, but I also do not volunteer the information because I do not want to seem like I am just attention-seeking and "chronically online".

Unsupportive experiences online

We found common themes among those who reported unsupportive experiences or a mix of positive and negative experiences on the internet as a trans person. Some confided that simply being online exposed them to more negativity than in their offline lives, with one person specifically commenting that being so online "messes you up."

Many reported facing transphobic harassment, dehumanization, and threats by existing as a trans person online. Exposure to transphobic slurs and

death threats led some to feel as though the internet had done more harm than good. For example, one person recounted, “It’s terrible. It’s all terrible. It’s so hard living like this; I get called slurs and told to die all the time just for being me.” Another commented:

I’m on the side of Twitter with all the *cool queer people*, so at least I have a parasocial wall of support, but, via retweets, quote tweets and replies, I also get to see the pure hatred of thousands of Americans every day. It’s discouraging...to see people like me being harassed, threatened, even killed so openly and no higher power coming to defend them. I sometimes try to think of this exposure as skin-hardening—maybe when I’m the one being harassed, I’ll have seen it enough times that I know how to handle it—but I don’t think I’ll ever get to a point of true unaffectedness, and in any case, I don’t think it’s healthy to get there anyway.

While many found support from within trans communities online, some reported negative experiences within these communities, describing cycles of gatekeeping identities, creating discourse, and obsessing over identity labels and passing, festering the internalized transphobia they already felt. For example, one person stated:

My time on the internet has helped shape my identity and expression in a very helpful way. However, it has also made me insecure in my identity. Trans people online can vary from being very accepting of other trans people to being incredibly critical about others’ experiences and the way they present themselves. I find that I still have to cope with internalized transphobia that I have picked up from online communities that possess...exclusionary attitudes.

A theme common amongst those who reported both supportive and unsupportive experiences online and unsupportive experiences offline was feeling lonely after spending time with online friends. Some felt they were too reliant on online communities for personal connections, often neglecting offline friendships in favor of online interactions. Others felt envious of those who had supportive family, friends, and communities offline, with one person commenting:

I think overall [being on the internet] has been good in giving me access to community and understanding I wouldn’t have otherwise, but at the same time I feel like there is even more loneliness in seeing people having true real-life connections and friends or even

just supportive family when the literal only person in my life is my spouse. I’m thankful to have them, but I feel like seeing the community I could have makes me miss the connection I could have.

General discussion

Our goal was to investigate the roles of support across multiple social contexts in the psychological adjustment of trans young adults. Across quantitative analyses, we found that support from offline social contexts was a better predictor of psychological adjustment than online support contexts. This is especially problematic, as 40.5% of our sample reported having no exclusively in-person friendships. Family support was the strongest predictors of psychological adjustment, and this is consistent with past literature (Russell & Fish, 2016; Snapp et al., 2015). Within the context of the risk and resilience framework, support from families and offline friends can serve as both a source of risk and resilience. Trans individuals are more likely than cisgender individuals to encounter systemic adversity that impacts “normative development” (risk), and in turn report higher levels of depression, suicidality, and other indicators of negative adjustment (Eisenberg et al., 2017; Johns et al., 2019). High levels of social support can provide resilience against risk, while low levels of social support can serve as a risk factor (Fergus & Zimmerman, 2005). The results of this study are aligned with past research, as more social support was associated with higher scores on measures of positive adjustment and lower scores on measures of negative adjustment. Family and offline friend support are crucial to positive adjustment in trans young adults.

We expected support from online friends to buffer trans youth from the negative impact of low in-person social support and to promote even better adjustment among youth with good in-person support. We found support for only the latter component of this hypothesis. Support from online friends served as a promotive factor, specifically, for the relationship between family support and pride. Past research has found that support from online friends and communities predicts positive adjustment *via* self-esteem, life satisfaction, and sense of community and fewer negative outcomes such as depression and

loneliness (Oh et al., 2014; Shaw & Gant, 2002). While our study found support for associations between online support and adjustment, the extent to which online support relates to psychological adjustment in comparison to support from offline social contexts is limited.

Support from online friends moderated the relationship between offline friend support and pride, with pride being highest among participants who were low in both online and offline friend support and lowest in those with high online and offline friend support. This is a counterintuitive finding, though the pride measure itself may contextualize these results. Example items from Testa et al.'s (2015) pride subscale of the Gender Minority Stress and Resilience Measure include, "My gender identity or expression makes me feel special or unique," and "I am like other people, but I am also special because my gender identity is different from my sex assigned at birth." Items like this highlight not only pride in one's identity, but also the perceived uniqueness of one's identity. If someone is heavily supported by LGBTQ friends, particularly if they have numerous connections who are also trans, then they may not feel that their identity is unique.

While this study did not find overwhelming evidence for the moderating role of online friend support, it did highlight the role of online support in certain indicators of psychological adjustment in trans young adults (pride and loneliness). This study also affirms the importance of offline support as an important source of risk and resilience (Craig et al., 2015; Selkie et al., 2020). Online support was significantly associated with some indicators of adjustment (pride, belongingness, positive well-being, and loneliness), but not others (self-esteem, satisfaction with life, internalized transphobia, depression, and anxiety). Online social contexts may provide support for trans young adults, but they may not be as meaningful in promoting resilience as offline social contexts (Ybarra et al., 2015). Future research should continue to clarify the role of online social support in the psychological adjustment of trans individuals.

A common theme in participants' open-ended responses is that people have highly individualized relationships and experiences with the internet, filtered by interactions and support from

both offline and online communities. Because these experiences are so individualized, it is impossible to generalize the internet as solely "good" or "bad" for psychological adjustment, instead representing a new social context that can be supportive, unsupportive, and in many cases, both supportive and unsupportive. For example, some who felt supported by online friends also expressed negative aspects to this support, as being online could also leave them vulnerable to hate and threats online or even become too reliant on the internet for their need to belong. While some people had positive experiences in online queer communities, others had negative experiences that led them to turn offline for support.

Because these online interactions are highly individualized and nuanced, it is difficult to identify specific themes of supportiveness and unsupportiveness from online sources. This may help to explain why the quantitative analyses found relatively little evidence of online friend support's influence on psychological adjustment. Online interactions can contain a multitude of supportive and unsupportive interactions, so there is a high level of individual-level variance that may bias the results, so that using online support as a predictor of adjustment is not sufficient to capture the essence of trans young adults' online experiences, as high levels of online support may also come at the cost of increased negative experiences online, offering a double-edged sword of support. If internet use can lead to a variety of positive and negative outcomes relevant to psychological adjustment, then the hypothesis of online support moderating offline support and adjustment is too simplistic to account for the nuances across varying relationships with the internet itself.

It is important to note the limitations of the qualitative analysis, as the question presented to participants did not specifically ask about the interaction between online and offline support, only about their experiences online. Because of this, sources of offline support were only discussed in the context of online experiences, limiting the scope of these findings.

Future research designs should account for the individualized and nuanced relationships trans

individuals have with the internet, which may not be simply quantified and analyzed in mass. Instead, it is important to adapt appropriate methodologies to account for both the benefits, drawbacks, and level of dependency on support from online friends. Additionally, future research could benefit from using measures of trans-specific social as opposed to general social support, which does not account for specific supportive behaviors such as affirming identity, helping with transition, and using proper names and pronouns. Because gender interacts with race, ethnicity, and culture in unique ways, findings that do not incorporate the interactions of these identity influences may not fully represent the processes of psychological adjustment (Bowleg et al., 2023).

Finally, it is important to note that all findings are in their historical context, reflecting a zeitgeist of heightened political tension centering trans identities. As reflected in the qualitative analysis, support from other trans people offered many a sense of hope for the future, where their identity and existence were not politicized, but normalized. While researchers are often critical of centering oppression in queer research (e.g., Levitt et al., 2023), it is important to take these findings in the context of their times and strive for a future where trans oppression is no longer a salient predictor of psychological adjustment, instead centering research on trans joy and liberation.

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Ethical approval

All procedures performed in this study were in accordance with the ethical standards of the American Psychological Association and the relevant university IRB.

Informed consent

Informed consent was obtained from all individual participants included in the study. Additional informed consent was obtained from all individual participants for anonymously reporting direct quotations from their responses.

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Data availability statement

Data are available upon request.

References

- Adkins, V., Masters, E., Shumer, D., & Selkie, E. (2018). Exploring transgender adolescents' use of social media for support and health information seeking. *Journal of Adolescent Health*, 62(2), S44. <https://doi.org/10.1016/j.jadohealth.2017.11.087>
- Aiken, L. S., & West, S. G. (1991). *Multiple regression: Testing and interpreting interactions*. Sage Publications, Inc.
- Alanko, K., & Lund, H. (2020). Transgender youth and social support: A survey study on the effects of good relationships on well-being and mental health. *YOUNG*, 28(2), 199–216. <https://doi.org/10.1177/1103308819850039>
- American Civil Liberties Union. (2023, December 21). *Mapping attacks on LGBTQ rights in U.S. state legislatures*. <https://www.aclu.org/legislative-attacks-on-lgbtq-rights>
- Arrington, E. G., & Wilson, M. N. (2000). A re-examination of risk and resilience during adolescence: Incorporating culture and diversity. *Journal of Child and Family Studies*, 9(2), 221–230. <https://doi.org/10.1023/A:1009423106045>
- Austin, A., Craig, S. L., Navega, N., & McInroy, L. B. (2020). It's my safe space: The life-saving role of the internet in the lives of transgender and gender diverse youth. *International Journal of Transgender Health*, 21(1), 33–44. <https://doi.org/10.1080/15532739.2019.1700202>
- Austin, A., Craig, S. L., D'Souza, S., & McInroy, L. B. (2022). Suicidality among transgender youth: Elucidating the role of interpersonal risk factors. *Journal of Interpersonal Violence*, 37(5–6), NP2696–NP2718. <https://doi.org/10.1177/0886260520915554>
- Başar, K., & Öz, G. (2016). Resilience in individuals with gender dysphoria: Association with perceived social support and discrimination. *Türk Psikiyatri Dergisi = Turkish Journal of Psychiatry*, 27(4), 225–234. <https://doi.org/10.5080/u17071>
- Başar, K., Öz, G., & Karakaya, J. (2016). Perceived discrimination, social support, and quality of life in gender dysphoria. *The Journal of Sexual Medicine*, 13(7), 1133–1141. <https://doi.org/10.1016/j.jsxm.2016.04.071>
- Bauer, G. R., Scheim, A. I., Pyne, J., Travers, R., & Hammond, R. (2015). Intervenable factors associated with suicide risk

- in transgender persons: A respondent driven sampling study in Ontario, Canada. *BMC Public Health*, 15(1), 525. <https://doi.org/10.1186/s12889-015-1867-2>
- Baumeister, R. F., & Leary, M. R. (1995). The need to belong: Desire for interpersonal attachments as a fundamental human motivation. *Psychological Bulletin*, 117(3), 497–529. <https://doi.org/10.1037/0033-2909.117.3.497>
- Beck, A. T., & Beck, R. W. (1972). Screening depressed patients in family practice: A rapid technique. *Postgraduate Medicine*, 52(6), 81–85. <https://doi.org/10.1080/00325481.1972.11713319>
- Bosse, J. D., Clark, K. D., Dion, K. A., & Chiodo, L. M. (2024). Transgender and nonbinary young adults' depression and suicidality is associated with sibling and parental acceptance-rejection. *Journal of Nursing Scholarship*, 56(1), 87–102. <https://doi.org/10.1111/jnu.12917>
- Bowleg, L., Malekzadeh, A. N., AuBuchon, K. E., Ghabrial, M. A., & Bauer, G. R. (2023). Rare exemplars and missed opportunities: Intersectionality within current sexual and gender diversity research and scholarship in psychology. *Current Opinion in Psychology*, 49, 101511. <https://doi.org/10.1016/j.copsyc.2022.101511>
- Boza, C., & Nicholson Perry, K. (2014). Gender-related victimization perceived social support, and predictors of depression among transgender Australians. *International Journal of Transgenderism*, 15(1), 35–52. <https://doi.org/10.1080/15532739.2014.890558>
- Catterall, J. S. (1998). Risk and resilience in student transitions to high school. *American Journal of Education*, 106(2), 302–333. <https://doi.org/10.1086/444184>
- Claes, L., Bouman, W. P., Witcomb, G., Thurston, M., Fernandez-Aranda, F., & Arcelus, J. (2015). Non-suicidal self-injury in trans people: Associations with psychological symptoms, victimization, interpersonal functioning, and perceived social support. *The Journal of Sexual Medicine*, 12(1), 168–179. <https://doi.org/10.1111/jsm.12711>
- Craig, S. L., Eaton, A. D., McInroy, L. B., Leung, V. W., & Krishnan, S. (2021). Can social media participation enhance LGBTQ+ youth well-being? Development of the social media benefits scale. *Social Media + Society*, 7(1), 161–179. <https://doi.org/10.1177/2056305121988931>
- Craig, S. L., McInroy, L., McCready, L. T., & Alaggia, R. (2015). Media: A catalyst for resilience in lesbian, gay, bisexual, transgender, and queer youth. *Journal of LGBT Youth*, 12(3), 254–275. <https://doi.org/10.1080/19361653.2015.1040193>
- Davey, A., Bouman, W. P., Arcelus, J., & Meyer, C. (2014). Social support and psychological well-being in gender dysphoria: A comparison of patients with matched controls. *The Journal of Sexual Medicine*, 11(12), 2976–2985. <https://doi.org/10.1111/jsm.12681>
- Diener, E. (1984). Subjective well-being. *Psychological Bulletin*, 95(3), 542–575. <https://doi.org/10.1037/0033-2909.95.3.542>
- Dowers, E., White, C., Cook, K., & Kingsley, J. (2020). Trans, gender diverse and non-binary adult experiences of social support: A systematic quantitative literature review. *International Journal of Transgender Health*, 21(3), 242–257. <https://doi.org/10.1080/26895269.2020.1771805>
- Eisenberg, M. E., Gower, A. L., McMorris, B. J., Rider, G. N., Shea, G., & Coleman, E. (2017). Risk and protective factors in the lives of transgender/gender nonconforming adolescents. *The Journal of Adolescent Health*, 61(4), 521–526. <https://doi.org/10.1016/j.jadohealth.2017.04.014>
- Evdokimova, T. (2023, July 10). *The global rise of anti-trans legislation*. Coda Story. <https://www.codastory.com/waronscience/lgbtq-trans-rights-2023/>
- Fergus, S., & Zimmerman, M. A. (2005). Adolescent resilience: A framework for understanding healthy development in the face of risk. *Annual Review of Public Health*, 26(1), 399–419. <https://doi.org/10.1146/annurev.publhealth.26.021304.144357>
- Hale, C. J., Hannum, J. W., & Espelage, D. L. (2005). Social support and physical health: The importance of belonging. *Journal of American College Health: J of ACH*, 53(6), 276–284. <https://doi.org/10.3200/JACH.53.6.276-284>
- Hefner, J., & Eisenberg, D. (2009). Social support and mental health among college students. *The American Journal of Orthopsychiatry*, 79(4), 491–499. <https://doi.org/10.1037/a0016918>
- International Lesbian, Gay, Bisexual, Trans, and Intersex Association. (2024, February 29). *Annual review of the human rights situation of lesbian, gay, bisexual, trans, and intersex people in Europe and Central Asia*. ILGA-Europe. https://www.ilga-europe.org/files/uploads/2024/02/2024_full_annual_review.pdf
- Johns, M. M., Lowry, R., Andrzejewski, J., Barrios, L. C., Demissie, Z., McManus, T., Rasberry, C. N., Robin, L., & Underwood, J. M. (2019). Transgender identity and experiences of violence victimization, substance use, suicide risk, and sexual risk behaviors among high school students—19 states and large urban school districts, 2017. *MMWR. Morbidity and Mortality Weekly Report*, 68(3), 67–71. <https://doi.org/10.15585/mmwr.mm6803a3>
- Johnson, A. H., & Rogers, B. A. (2020). “We’re the normal ones here”: Community involvement, peer support, and transgender mental health. *Sociological Inquiry*, 90(2), 271–292. <https://doi.org/10.1111/soin.12347>
- Kia, H., MacKinnon, K. R., Abramovich, A., & Bonato, S. (2021). Peer support as a protective factor against suicide in trans populations: A scoping review. *Social Science & Medicine*, 279, 114026. <https://doi.org/10.1016/j.socscimed.2021.114026>
- Levitt, H. M., Kehoe, K. A., & Hand, A. B. (2023). Beyond minority stress: Toward a multidimensional psychology of trans/nonbinary gender. *Current Opinion in Psychology*, 49, 101515. <https://doi.org/10.1016/j.copsyc.2022.101515>
- Luthar, S. S., Cicchetti, D., & Becker, B. (2000). The construct of resilience: A critical evaluation and guidelines for future work. *Child Development*, 71(3), 543–562. <https://doi.org/10.1111/1467-8624.00164>
- Malone, G. P., Pillow, D. R., & Osman, A. (2012). The general belongingness scale (GBS): Assessing achieved

- belongingness. *Personality and Individual Differences*, 52(3), 311–316. <https://doi.org/10.1016/j.paid.2011.10.027>
- Masten, A. S. (2001). Ordinary magic: Resilience processes in development. *The American Psychologist*, 56(3), 227–238. <https://doi.org/10.1037/0003-066x.56.3.227>
- McConnell, E. A., Birkett, M., & Mustanski, B. (2016). Families matter: Social support and mental health trajectories among lesbian, gay, bisexual, and transgender youth. *The Journal of Adolescent Health*, 59(6), 674–680. <https://doi.org/10.1016/j.jadohealth.2016.07.026>
- Milton, D. C., & Knutson, D. (2023). Family of origin, not chosen family, predicts psychological health in a LGBTQ+ sample. *Psychology of Sexual Orientation and Gender Diversity*, 10(2), 269–278. <https://doi.org/10.1037/sgd0000531>
- Moran, T. E., Chen, C. Y. C., & Tryon, G. S. (2018). Bully victimization, depression, and the role of protective factors among college LGBTQ students. *Journal of Community Psychology*, 46(7), 871–884. <https://doi.org/10.1002/jcop.21978>
- Oh, H. J., Ozkaya, E., & LaRose, R. (2014). How does online social networking enhance life satisfaction? The relationships among online supportive interaction, affect, perceived social support, sense of community, and life satisfaction. *Computers in Human Behavior*, 30, 69–78. <https://doi.org/10.1016/j.chb.2013.07.053>
- Olson, K. R., Durwood, L., DeMeules, M., & McLaughlin, K. A. (2016). Mental health of transgender children who are supported in their identities. *Pediatrics*, 137(3), e20153223. <https://doi.org/10.1542/peds.2015-3223>
- Rosenberg, M. (1965). *Society and the adolescent self-image*. Princeton University Press. <https://psycnet.apa.org/doi/10.1037/t01038-000>
- Rothbaum, B., Etengoff, C., & Uribe, E. (2022). Transgender community resilience on YouTube: Constructing an informational, emotional, and sociorelational support exchange. *Journal of Community Psychology*, 50(5), 2366–2384. <https://doi.org/10.1002/jcop.22781>
- Russell, D., Peplau, L. A., & Ferguson, M. L. (1978). Developing a measure of loneliness. *Journal of Personality Assessment*, 42(3), 290–294. https://doi.org/10.1207/s15327752jpa4203_11
- Russell, S. T., & Fish, J. N. (2016). Mental health in lesbian, gay, bisexual, and transgender (LGBT) youth. *Annual Review of Clinical Psychology*, 12(1), 465–487. <https://doi.org/10.1146/annurev-clinpsy-021815-093153>
- Ryan, C., Russell, S. T., Huebner, D., Diaz, R., & Sanchez, J. (2010). Family acceptance in adolescence and the health of LGBT young adults. *Journal of Child and Adolescent Psychiatric Nursing*, 23(4), 205–213. <https://doi.org/10.1111/j.1744-6171.2010.00246.x>
- Scandurra, C., Amodeo, A. L., Valerio, P., Bochicchio, V., & Frost, D. M. (2017). Minority stress, resilience, and mental health: A study of Italian transgender people. *Journal of Social Issues*, 73(3), 563–585. <https://doi.org/10.1111/josi.12232>
- Selkie, E., Adkins, V., Masters, E., Bajpai, A., & Shumer, D. (2020). Transgender adolescents' uses of social media for social support. *The Journal of Adolescent Health*, 66(3), 275–280. <https://doi.org/10.1016/j.jadohealth.2019.08.011>
- Simons, L., Schrager, S. M., Clark, L. F., Belzer, M., & Olson, J. (2013). Parental support and mental health among transgender adolescents. *The Journal of Adolescent Health*, 53(6), 791–793. <https://doi.org/10.1016/j.jadohealth.2013.07.019>
- Singh, A. A., Hays, D. G., & Watson, L. S. (2011). Strength in the face of adversity: Resilience strategies of transgender individuals. *Journal of Counseling & Development*, 89(1), 20–27. <https://doi.org/10.1002/j.1556-6678.2011.tb00057.x>
- Singh, A. A., & McKleroy, V. S. (2011). “Just getting out of bed is a revolutionary act” the resilience of transgender people of color who have survived traumatic life events. *Traumatology*, 17(2), 34–44. <https://doi.org/10.1177/1534765610369261>
- Singh, A. A., Meng, S. E., & Hansen, A. W. (2014). “I am my own gender”: Resilience strategies of trans youth. *Journal of Counseling & Development*, 92(2), 208–218. <https://doi.org/10.1002/j.1556-6676.2014.00150.x>
- Shaw, L. H., & Gant, L. M. (2002). In defense of the Internet: The relationship between Internet communication and depression, loneliness, self-esteem, and perceived social support. *Cyberpsychology & Behavior*, 5(2), 157–171. <https://doi.org/10.1089/109493102753770552>
- Snapp, S. D., Watson, R. J., Russell, S. T., Diaz, R. M., & Ryan, C. (2015). Social support networks for LGBT young adults: Low cost strategies for positive adjustment. *Family Relations*, 64(3), 420–430. <https://doi.org/10.1111/fare.12124>
- Southwick, S. M., Bonanno, G. A., Masten, A. S., Panter-Brick, C., & Yehuda, R. (2014). Resilience definitions, theory, and challenges: Interdisciplinary perspectives. *European Journal of Psychotraumatology*, 5(1), 25338. <https://doi.org/10.3402/ejpt.v5.25338@zept20.2014.5.issue-s4>
- Steinke, J., Root-Bowman, M., Estabrook, S., Levine, D. S., & Kantor, L. M. (2017). Meeting the needs of sexual and gender minority youth: Formative research on potential digital health interventions. *The Journal of Adolescent Health*, 60(5), 541–548. <https://doi.org/10.1016/j.jadohealth.2016.11.023>
- Stieglitz, K. A. (2010). Development, risk, and resilience of transgender youth. *The Journal of the Association of Nurses in AIDS Care: JANAC*, 21(3), 192–206. <https://doi.org/10.1016/j.jana.2009.08.004>
- Tankersley, A. P., Gafsky, E. L., Dike, J., & Jones, R. T. (2021). Risk and resilience factors for mental health among transgender and gender nonconforming (TGNC) youth: A systematic review. *Clinical Child and Family Psychology Review*, 24(2), 183–206. <https://doi.org/10.1007/s10567-021-00344-6>
- Testa, R. J., Habarth, J., Peta, J., Balsam, K., & Bockting, W. (2015). Development of the gender minority stress and resilience measure. *Psychology of Sexual Orientation and Gender Diversity*, 2(1), 65–77. <https://doi.org/10.1037/sgd0000081>
- Testa, R. J., Jimenez, C. L., & Rankin, S. (. (2014). Risk and resilience during transgender identity development: The effects of awareness and engagement with other transgender people on affect. *Journal of Gay & Lesbian Mental Health*, 18(1), 31–46. <https://doi.org/10.1080/19359705.2013.805177>

- Topp, C. W., Østergaard, S. D., Søndergaard, S., & Bech, P. (2015). The WHO-5 Well-Being Index: A systematic review of the literature. *Psychotherapy and Psychosomatics*, 84(3), 167–176. <https://doi.org/10.1159/000376585>
- The Trevor Project. (2023, January). *Issues impacting LGBTQ youth*. The Trevor Project. https://www.thetrevorproject.org/wp-content/uploads/2023/01/Issues-Impacting-LGBTQ-Youth-Morning-Consult-Poll-Jan-2023_Public.pdf
- Veale, J. F., Watson, R. J., Peter, T., & Saewyc, E. M. (2017). Mental health disparities among Canadian transgender youth. *The Journal of Adolescent Health: Official Publication of the Society for Adolescent Medicine*, 60(1), 44–49. <https://doi.org/10.1016/j.jadohealth.2016.09.014>
- Williamson, M. (2024). A global analysis of transgender rights: Introducing the Trans Rights Indicator Project (TRIP). *Perspectives on Politics*, 22(3), 799–818. <https://doi.org/10.1017/S1537592723002827>
- Wilson, E. C., Iverson, E., Garofalo, R., & Belzer, M. (2012). Parental support and condom use among transgender female youth. *The Journal of the Association of Nurses in AIDS Care*, 23(4), 306–317. <https://doi.org/10.1016/j.jana.2011.09.001>
- Ybarra, M. L., Mitchell, K. J., Palmer, N. A., & Reisner, S. L. (2015). Online social support as a buffer against online and offline peer and sexual victimization among US LGBT and non-LGBT youth. *Child Abuse & Neglect*, 39, 123–136. <https://doi.org/10.1016/j.chiabu.2014.08.006>
- Zimet, G. D., Dahlem, N. W., Zimet, S. G., & Farley, G. K. (1988). The multidimensional scale of perceived social support. *Journal of Personality Assessment*, 52(1), 30–41. https://doi.org/10.1207/s15327752jpa5201_2
- Zsido, A. N., Teleki, S. A., Csokasi, K., Rozsa, S., & Bandi, S. A. (2020). Development of the short version of the Spielberger state-trait anxiety inventory. *Psychiatry Research*, 291, 113223. <https://doi.org/10.1016/j.psychres.2020.113223>